

Shanmukha Roopesh

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PROFESSIONAL SUMMARY

Motivated and detail-oriented MCA student with a strong foundation in Java Full Stack Development and a passion for building scalable, secure web applications. Proficient in Spring Boot, ReactJS, Microservices, and RESTful API design, with hands-on experience in data structures, algorithms, and problem-solving using Java. Seeking a DSE or SP role at Infosys to contribute technical expertise, engineering discipline, and a strong drive to deliver high-quality software solutions in a collaborative environment.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL

DSA & Problem Solving: Arrays, Linked Lists, Binary Trees, Graphs, Dynamic Programming, Recursion, Sorting & Searching Algorithms

Backend: Spring Boot, Spring Security, JWT, Hibernate, JDBC, Microservices, REST APIs

Frontend: ReactJS, HTML5, CSS3, JavaScript (ES6+)

Databases: MySQL, JDBC

Tools & DevOps: Git, GitHub, Postman, Jenkins (CI/CD), Docker, Kubernetes

Web Server: Apache Tomcat

PROJECTS

Secure Data Sharing and Authorized Searchable Framework for E-Healthcare System | Tech: Java (J2EE), Spring Boot, MySQL, JDBC, HTML, CSS, JavaScript, Apache Tomcat

- Developed a secure e-healthcare platform enabling privacy-preserving sharing of patient health records (PHR) over cloud infrastructure.
- Implemented Conditional Proxy Re-Encryption (CPRE) to enforce controlled, authorized access to sensitive medical data by verified users.
- Applied searchable encryption techniques to allow efficient querying on encrypted records without exposing plaintext data.
- Integrated Spring Security and JWT-based authentication to manage role-based access control for patients, doctors, and administrators.
- Designed a responsive and accessible UI using HTML, CSS, and JavaScript for seamless interaction between patients and healthcare providers.
- Deployed the application on Apache Tomcat with MySQL as the backend database, achieving consistent sub-300ms query response times.

AlgoBoard - DSA Visualizer and Problem Tracker (Full Stack) | Tech: Java, Spring Boot, ReactJS, MySQL, REST API, Graph Algorithms, Dynamic Programming

- Built a full-stack web application to help students track, practice, and visualize core DSA problems across topics including Trees, Graphs, DP, and Sorting.
- Developed a Spring Boot REST API backend with endpoints for problem CRUD operations, progress tracking, and difficulty-based filtering, secured with JWT authentication.
- Implemented server-side graph traversal engines (BFS and DFS) in Java returning step-by-step traversal sequences as JSON for frontend visualization.
- Designed Dynamic Programming modules for classic problems (Longest Common Subsequence, 0/1 Knapsack) with memoization, reducing time complexity from $O(2^n)$ to $O(n*m)$.
- Built an interactive ReactJS frontend with real-time progress dashboards, topic-wise completion stats, and a visual tree renderer using recursive component rendering.
- Integrated Jenkins CI/CD pipeline with GitHub hooks for automated build, test, and deployment to a Dockerized environment on every pull request merge.
- Wrote 80+ JUnit test cases covering service layer logic, graph traversal correctness, and DP output validation with 90%+ code coverage.

EDUCATION

Master of Computer Applications (MCA) - Krishna Chaitanya Institute of Science and Technology | CGPA: 7.8

Bachelor of Computer Applications (BCA) - Krishna Chaitanya Degree College | CGPA: 7.89

Intermediate (MPC) - Narayana Junior College | CGPA: 6.0

SSC - Sri Netaji MSR Pilot High School | CGPA: 8.2

CERTIFICATIONS

- Java Full Stack Development - Frontlines Edutech
- Frontend Development - Udemy

DECLARATION

I hereby declare that the above information is true and correct to the best of my knowledge and belief.